

1. To write a Python program to add two numbers entered by the user.

```
A = int(input("Enter first number: "))
B = int(input("Enter second number: "))
Add=A+B
print(Add)
```

```
Enter first number: 5
Enter second number: 3
8
```

2. To write a Python program to interchange the values of two variables.

```
a = int(input("Enter first number: "))
b = int(input("Enter second number: "))
temp = a
a = b
b = temp
print("After swapping:")
print("First number =", a)
print("Second number =", b)
```

```
Enter first number: 10
Enter second number: 20
After swapping:
First number = 20
Second number = 10
```

3. To write a python program to find the largest among three using conditional statement.

```
a = int(input("Enter first number: "))
b = int(input("Enter second number: "))
c = int(input("Enter third number: "))
if a >= b and a >= c:
    print("Largest number is", a)
elif b >= a and b >= c:
    print("Largest number is", b)
else:
    print("Largest number is", c)
```

```
Enter first number: 11
Enter second number: 22
Enter third number: 9
Largest number is 22
```

4. To write a Python program to check whether a given number is even or odd.

```
n = int(input("Enter a number: "))
if n % 2 == 0:
    print("Even number")
else:
    print("Odd number")
```

```
Enter a number: 22
Even number
```

5. To write a Python program to determine whether a number is positive, negative, or zero.

```
n = int(input("Enter a number: "))
if n > 0:
    print("Positive number")
elif n < 0:
    print("Negative number")
else:
    print("Zero")
```

```
Enter a number: 5
Positive number
```

6. To write a Python program to check whether a given year is a leap year.

```
year = int(input("Enter year: "))
if (year % 400 == 0) or (year % 4 == 0 and year % 100 != 0):
    print("Leap year")
```

```
else:  
    print("Not a leap year")
```

Enter year: 2026
Not a leap year

7. To write a Python program to perform basic arithmetic operations using operators.

```
a = int(input("Enter first number: "))  
b = int(input("Enter second number: "))  
print("Addition =", a + b)  
print("Subtraction =", a - b)  
print("Multiplication =", a * b)  
print("Division =", a / b)
```

Enter first number: 2
Enter second number: 9
Addition = 11
Subtraction = -7
Multiplication = 18
Division = 0.2222222222222222

8. To write a Python program to display the multiplication table of a given number.

```
n = int(input("Enter a number: "))  
for i in range(1, 11):  
    print(n, "x", i, "=", n * i)
```

Enter a number: 9
9 x 1 = 9
9 x 2 = 18
9 x 3 = 27
9 x 4 = 36
9 x 5 = 45
9 x 6 = 54
9 x 7 = 63
9 x 8 = 72
9 x 9 = 81
9 x 10 = 90

9. To write a Python program to calculate the factorial of a number using a loop.

```
n = int(input("Enter a number: "))  
fact = 1  
for i in range(1, n + 1):  
    fact = fact * i  
  
print("Factorial =", fact)
```

Enter a number: 5
Factorial = 120

10. To write a Python program to find the sum of the first N natural numbers.

```
N = int(input("Enter value of N: "))  
sum = 0  
for i in range(1, N + 1):  
    sum = sum + i  
print("Sum =", total)
```

Enter value of N: 15
Sum = 120

11. To write a Python program to print numbers from 1 to N using a for loop.

```
N = int(input("Enter value of N: "))  
  
for i in range(1, N + 1):  
    print(i)
```

Enter value of N: 10
1
2
3
4
5
6
7

```
8
9
10
```

12. To write a Python program to print numbers in reverse order from N to 1.

```
n = int(input("Enter value of N: "))
for i in range(n, 0, -1):
    print(i)
```

```
Enter value of N: 9
9
8
7
6
5
4
3
2
1
```

13. To write a Python program to print all even numbers up to a given limit.

```
limit = int(input("Enter the limit: "))
for i in range(1, limit+1):
    if i%2==0:
        print(i)
```

```
Enter the limit: 12
2
4
6
8
10
12
```

14. To write a Python program to reverse the digits of a given number.

```
n = int(input("Enter a number: "))

rev = 0

while n > 0:
    digit = n % 10
    rev = rev * 10 + digit
    n = n // 10

print("Reverse =", rev)
```

```
Enter a number: 29
Reverse = 92
```

15. To write a Python program to count the number of digits in a given integer.

```
num = int(input("Enter a number: "))
temp=abs(num)
count = 0
if temp==0:
    count=1
while temp > 0:
    count = count + 1
    temp = temp // 10
print("Number of digits =", count)
```

```
Enter a number: 365
Number of digits = 3
```

16. To write a Python program to calculate the sum of digits of a given number.

```
n = int(input("Enter a number: "))
total = 0
while n > 0:
    digit = n % 10
    total = total + digit
    n = n // 10
print("Sum of digits =", total)
```

```
Enter a number: 36
Sum of digits = 9
```

17. To write a Python program to check whether a given number is prime.

```
n = int(input("Enter a number: "))
count = 0
for i in range(1, n + 1):
    if n % i == 0:
        count = count + 1
if count == 2:
    print("Prime number")
else:
    print("Not a prime number")
```

```
Enter a number: 5
Prime number
```

18. To write a Python program to generate the Fibonacci series up to N terms.

```
n = int(input("Enter number of terms: "))
a = 0
b = 1
for i in range(n):
    print(a, end=" ")
    c = a + b
    a = b
    b = c
```

```
Enter number of terms: 7
0 1 1 2 3 5 8
```

19. To write a Python program to check whether a string is a palindrome.

```
text = input("Enter a string: ")
if text == text[::-1]:
    print("Palindrome")
else:
    print("Not a palindrome")
```

```
Enter a string: malayalam
Palindrome
```

20. To write a Python program to check whether a given number is an Armstrong number.

```
n = int(input("Enter a number: "))
temp = n
total = 0
digits = len(str(n))

while temp > 0:
    digit = temp % 10
    total = total + digit ** digits
    temp = temp // 10
if total == n:
    print("Armstrong number")
else:
    print("Not an Armstrong number")
```

```
Enter a number: 153
Armstrong number
```

